Computer Science & IT



C Programming

Control Flow Statements

Lecture No. 03



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Recap of Previous Lecture



Topic If else

Topic If else if

Topic Switch

Topic

Topic

Slide

Topics to be Covered



Topic Switch

Topic Loops

Topic

Topic

Topic

Slide

Break Statement transfer control from Loop & Switch out side of the block in C Language.

defaut is ~~th~~. a Switch case executed when No case is satisfied

Switch & dfout are Not mandatory.

Switch (Expression) {

Expression

- Arithmetic Expression
- Relational Expression
- Logical Expression

}

after evaluating the expression the value will be tested for **equality** against list of value

Each List value is called a **Case**

```
switch ( Expression ) {
    case Constant Expression :
        Statement; break;
    case constant expression :
        Statement; break;
    :
    defaut : Statements; break;
}
```

Constant Numeric

Application

Menu

Stack : 1. push
2. pop
3. Initialize
4. Exit

Menu creation Data structure Stack

Cases

→ 1. Push

→ 2. POP

→ 3. Initialization

→ 4 Exit

default:

If No case Mathes then default is executed

*

No default is Not mandetory.

break;

```
int a = 5
Switch (a+1) {
Case 1 : printf("2");
Case 6 : printf("3");
Case 7 : printf("4");
default : printf("5");
}
```

6

break;

3, 4, 5

if

break;

present

3,4

if break is Not present then all subsequent case evaluates to true:



Question

```
#include <stdio.h>
    int main(){
     int a = 5;
      switch(a){
        case 4: printf("%d",5);
        case 5: printf("%d",5);
         case 6: printf("%d",6);
      }
    }
```

Numeric Constant

Output

5 6

Reason

No break in case 5
Hence case 6 will be evaluated true.

Slide



switch(expression)

```
#include <stdio.h>
    int main(){
     int a = 5;
      switch(a-1){
            case 4: printf("%d",5);
             case 5: printf("%d",5);
             case 6: printf("%d",6);
      }
    }
```

4

4-1

5-1 = 4

Output

556

Reason

Case 4 is true and No break after this

Slide



switch(expression)

$13/4 = 3$ ↑ expression

```
#include <stdio.h>
int main(){
        switch(13/4){
        case 3: printf("%d",4); break;
        case 4: printf("%d",2); break;
        case 5: printf("%d",5); break;
        }
    }
```

Every case is having break is true break is ☞

Output

4

Reason :

expression is evaluated to 3. ☞ case 3

Slide



switch(expression)

```c
#include <stdio.h>
int main(){
        switch(13/4.0){
        case 3: printf("%d",4); break ;
        case 4: printf("%d",2); break ;
        case 5: printf("%d",5); break ;
        }
    }
```

13/4.0 ←

float Constant

Output

error

Reason

Expression evaluated to float.

Slide



switch(expression)

```
#include <stdio.h>
int main(){
      switch(printf("ABCD")){
         case 4: printf("%d",4); break ;
         case 3: printf("%d",2); break ;
         case 5: printf("%d",5); break ;
}
}
```

ABCD

4

4

Switch(4) {

4 characters are printed

Output

ABCD4

Reason

printf statement return No. of character printed

Slide

$X = -2024$

$\sim (X = X + 5)$

$X = -2019$

-2024

$-2024 + 5$

-2019

$2048 \quad -2048$

$2019 + 29$

$\begin{array}{r} {\scriptstyle 1} \\ 29 \end{array}$

$\begin{array}{r} {\scriptstyle 1} \\ -2019 \end{array}$

1 1 1 1 0 0 0 0 0 0 1 1 1 0 1 1
0 0 0 0 1 1 1 1 1 1 0 0 0 1 0

printf function returns No. of character printed (Space is also counted)



switch(expression)

Character → integer

integer

```
#include <stdio.h>
int main(){
    switch('d'){
    case 'a': printf("%d",4); break ;
    case 'b': printf("%d",2); break ;
    case 'c': printf("%d",5); break ;
    default: print("None");
}
```

character constant

a - 97
b - 98
c - 99
d - 100

Output

None

Reason

'd' is a integer

ASCII value

Case constant value

ASCII

Slide

* Switch (expression)

→ can be a character Constant allowed

Case can also have character constant



switch(expression)

```
#include <stdio.h>
int main(){
        switch('d'){
          case 'a': printf("%d",4); break ;
          case 'b': printf("%d",2); break ;
          default: print("None");
          case 'c': printf("%d",5); break ;

}
```

Output

None5

Reason

default position is Not fixed. and No break;

Slide

Its ~~Nont~~ Not mandetory that the default is last switch case.



switch(expression)

HW

Output

```
#include <stdio.h>
int main(){
        switch(4){
        case 2: printf("%d",4); break ;
        case 1+1: printf("%d",2); break ;
        default: printf("None");
    }
}
```

Reason

Slide



switch(expression)

```
#include <stdio.h>
int main(){
        int a=2;
         switch(4){
         case 2: printf("%d",4); break ;
         case a: printf("%d",2); break ;
         default: printf("None");
      }
}
```

Output

Reason



GATE 2012

Choice ~~Choice~~ Choice A Choic B No choice

What will be the output of the following C program segment?

```
Char inChar = 'A';
switch (inChar ) {
case 'A' : printf ("Choice A\n");
     case 'B' :
     case 'C' : printf("Choice B");
     case 'D' :
     case 'E' :
     default  : printf ("No Choice") ; }
```

New Line

(A) No choice
(B) Choice A
(C) Choice A
Choice B No choice
(D) Program gives no output as it is erroneous

Choice A
Choic BNo choice

Slide



What is Iterative Statement ?

Control Statement Iterative Statement.

Loop : If we want to repeat a process / computation again & again.

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Student grade can be decided.

Suppose we need to decide grade of 100 students and print the mark sheet.

Repitition are there

```
stude 1 - grade - print;
Student 2 - grade print;
:
Student 100 - grade print.
```

To perform repetative task each language is having provision of loops.

Loop :
* for loop
* while loop
* do while



For loop: Syntax

Print My name 100 times

```
for ( expr1 ; expr2 ; expr3 ) {

}
```

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expr$_1$ is called initialization and executed one and at the begining.

Iteration caunt begining decided by expr$_1$.

$expr_2$ is Condional expression.

if condition evaluates to true the

for loop block will be executed

Every time condition is true block will

exeute if false ~~for~~ block will Not execute.

expr3 : iteration statement Iterative count

Increment / Decrement.

expr3 is evaluated after all statement of block is executed.

```
for( i=1; i<=100; i++) {
    printf("I am a good student");
}
after printf
after Last stmt of block
```

i=1	Iteration 1	1<=100	- print ✓
i=2;	Iteration 2	2<=100	print ✓
i=3	Iteration 3	3<=100	print ✓
i=100	Iteration 100	100<=100	print ✓
i=101	Iteration 101	101<=100	false block will Not execute



For loop: Syntax

Different range of numbers

```
for(i =1;i<=10;i++)
        printf("Name")
```



For loop: Syntax

Different range of numbers

```
for(i =-10; i<=21; i++)
        printf("Name")
```



For loop: Syntax

Different range of numbers

```
for(i =-10; i<21; i++)
      printf("Name")
```



For loop: Syntax

Different range of numbers

```
for(i =100; i>=1; i++)
        printf("Name")
```



Question

```
#include <stdio.h>
int main(){
    int i=-1;
    for(++i;i++;i++)
        printf("Name");
    return 0;
}
```

Number of times loop is Executed

(A) 0

(B) 1

(C) Infinite

(D) 2

Slide



GATE 2015

```
#include<stdio.h>
int main()
{
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k-=--j;
    for (i=0; i<5; i++)
    {
        switch(i+k)
        {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
    return 0;
}
```

t.me/Abhishekshaarmapw



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Slide

THANK - YOU